

SAT Mathematics

SAT մաթեմատիկա

COURSE CODE	MTH-SAT
PROGRAM	Advanced Placement and Examination Preparation · Առաջադեմ և քննական պատրաստություն
LEVEL	Typically Grades 10–12 · digital adaptive SAT
DURATION	48–60 instructional hours (approx. 40 lessons) plus practice testing
PREREQUISITE	Algebra I and Geometry completed; Algebra II completed or in progress.
LEADS TO	Continued practice testing, or ACT Mathematics (MTH-ACT) if the student is comparing formats

COURSE DESCRIPTION

Դասընթացի նկարագրություն

This course prepares a student for the mathematics section of the digital SAT. The digital SAT Math section is 70 minutes long across two 22-question modules, and is section-adaptive: performance on the first module determines whether the second is the harder or the easier form, and reaching the harder form is necessary for a high score. A calculator — either the student's own approved device or the built-in Desmos graphing calculator — is permitted throughout. The course covers all four tested content domains alongside the strategic skills the adaptive format specifically rewards.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

SAT preparation done badly is a term of timed practice tests with no diagnosis attached. Վարժարան structures it the other way around: every practice section is followed by an item-level analysis separating content gaps from strategic errors from timing errors, and the following lessons are assigned from that analysis rather than from a fixed sequence. This matters more on the digital SAT than it did on paper, because the adaptive format concentrates the scoring consequence in the first module — a student who mismanages time early is routed to the easier second module and capped regardless of what they know. Teachers also train Desmos deliberately: on the digital SAT it is frequently faster to graph than to solve, and students who do not know this leave points on the table.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Solves linear equations and systems reliably	Algebra is the largest tested domain; weakness here caps the achievable score.
Works with quadratics, exponents, and function notation	The Advanced Math domain; typically the differentiator between mid and high scores.
Interprets graphs, tables, ratios, and percentages	Problem-Solving and Data Analysis; heavily contextual and reading-dependent.
Reads a dense word problem accurately under time pressure	The most common non-content cause of lost points, and rarely diagnosed as such.
Has taken at least one full official practice test	Provides the baseline that the entire course is sequenced from.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Solve linear equations, inequalities, systems, and functions fluently and quickly.
- 2 Interpret linear relationships in context and translate between words, equations, tables, and graphs.
- 3 Work with quadratic, exponential, polynomial, radical, and rational expressions and equations.
- 4 Analyze nonlinear functions and their graphs, including transformations and key features.
- 5 Solve ratio, rate, proportion, percentage, and unit conversion problems accurately.
- 6 Interpret data from tables and graphs, and work with probability, statistics, and inference.
- 7 Apply geometry and trigonometry: area, volume, angles, triangles, circles, and right triangle trigonometry.
- 8 Use the built-in Desmos calculator strategically for graphing, solving, and verification.
- 9 Manage time across two adaptive modules, including deliberate triage and flagging.
- 10 Diagnose one's own errors by category and direct practice accordingly.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 5-6 LESSONS

Diagnostic, Format Strategy, and Desmos

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TOPICS COVERED

- The digital SAT format: two modules, 22 questions each, 35 minutes each
- How section adaptivity works and why the first module matters most
- Multiple-choice and student-produced response (grid-in) question types
- Entering student-produced responses correctly, including fractions and negatives
- The reference sheet: what is provided and what must be known
- Full diagnostic test and item-level error analysis
- Classifying errors: content gap, strategic error, timing error, careless error
- The Desmos built-in calculator: graphing to find intersections and zeros
- Desmos for systems, inequalities, regressions, and statistics
- When graphing is faster than algebra, and when it is not
- Flagging, triage, and the review screen
- Building a personal score plan from the diagnostic

LEARNING OBJECTIVES

- 1 Explain how the adaptive format affects scoring and adjust pacing accordingly.
- 2 Classify every diagnostic error into one of four categories.
- 3 Use Desmos to solve at least five distinct question types faster than by algebra.
- 4 Enter student-produced responses in valid formats without losing points to entry errors.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Classifies own errors accurately in 9 of 10 items when compared with the teacher's classification.
- Solves designated question types with Desmos in under half the algebraic time.
- Produces a written personal score plan naming three priority domains.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The adaptive format is believed to make early questions less important.	The opposite is true: the first module determines which second module is delivered and therefore the score ceiling. Pace the first module deliberately.
Desmos is used only as a numerical calculator.	Its graphing capability is the advantage. Practice graphing-first solutions until they are the default for the relevant question types.

MISCONCEPTION	RECOMMENDED RESPONSE
Grid-in answers are entered in unaccepted formats.	Practice the entry rules explicitly. Losing a point to formatting after solving correctly is the most avoidable loss on the test.

TEACHING NOTE

The error classification introduced here drives the entire course. A student whose errors are 60% timing does not need more content teaching, and treating them identically to a content-gap student wastes the whole term.

UNIT 2 • 10-11 LESSONS • THE LARGEST TESTED DOMAIN**Algebra**

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TOPICS COVERED

- Linear equations in one and two variables
- Linear functions and interpreting slope and intercept in context
- Systems of two linear equations and the number of solutions
- Linear inequalities in one and two variables
- Systems of linear inequalities
- Building linear models from verbal descriptions
- Literal equations and rearranging formulas
- Absolute value equations and inequalities
- Interpreting equations that model real contexts
- Recognizing the SAT's recurring algebra question templates
- Solving by strategic substitution and by back-solving from the answer choices
- Choosing between algebra, graphing, and answer testing

LEARNING OBJECTIVES

- 1 Solve any tested linear equation, system, or inequality accurately in under 90 seconds.
- 2 Translate a verbal description into a linear model and interpret its parameters.
- 3 Determine the number of solutions of a system from the equations alone.
- 4 Select the fastest available method — algebra, graphing, or back-solving — for a given item.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers algebra domain items at 90% accuracy within the time budget.
- Selects the fastest method in 8 of 10 items when compared with the teacher's choice.
- Translates verbal descriptions into models correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Every problem is solved algebraically because that is how school taught it.	The SAT rewards the fastest correct route. Back-solving and graphing are often faster; practice them as first-line methods, not fallbacks.

MISCONCEPTION	RECOMMENDED RESPONSE
The question asked is not the quantity solved for.	Underline what is asked before starting. Solving for x when $2x + 1$ was requested is the most common careless error on the section.
A system's special cases are approached by full solution.	Compare coefficient ratios. Recognition is faster than solving.

TEACHING NOTE

Algebra is the largest domain and the most learnable. A student weak here should spend disproportionate time in this unit — the return per hour is higher than anywhere else on the test.

UNIT 3 • 10-11 LESSONS**Advanced Mathematics**

Առաջադեմ մաթեմատիկա

TOPICS COVERED

- Quadratic equations: factoring, the formula, completing the square, and graphing
- The discriminant and the number of real solutions
- Vertex form and identifying maxima and minima
- Nonlinear systems and their intersections
- Polynomial expressions, factoring, and the relationship between factors and zeros
- Exponential functions, growth and decay, and their parameters
- Rational expressions and equations, including extraneous solutions
- Radical expressions and equations
- Exponent rules including rational and negative exponents
- Function notation, evaluation, composition, and transformations
- Interpreting nonlinear models in context
- Equivalent expressions and structural rewriting

LEARNING OBJECTIVES

- 1** Solve quadratic and nonlinear equations by the fastest available method.
- 2** Interpret the parameters of a nonlinear model in context.
- 3** Recognize equivalent forms of an expression and choose the most revealing one.
- 4** Use function notation and transformations fluently under time pressure.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers advanced math items at 88% accuracy within the time budget.
- Identifies equivalent expressions correctly in 9 of 10 trials.
- Interprets nonlinear model parameters in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The quadratic formula is used where factoring or graphing would be far faster.	Check for factorability and consider graphing first. Method selection is a trained skill and it is worth minutes.
Extraneous solutions are not checked in radical and rational equations.	The SAT includes them deliberately. Check every solution in the original equation.
Structural rewriting questions are approached by full expansion.	Look for the intended structure — difference of squares, common factor, completed square. Expansion usually loses time.

TEACHING NOTE

This domain most sharply distinguishes scores above and below roughly 650. Students targeting a high score should expect to spend the largest share of their independent practice here.

UNIT 4 • 8-9 LESSONS**Problem-Solving and Data Analysis**

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TOPICS COVERED

- Ratios, rates, and proportional relationships
- Unit conversion and unit analysis in multi-step problems
- Percentages, percent change, and successive changes
- Reading tables, bar graphs, line graphs, scatter plots, and histograms
- Two-way tables and conditional relative frequency
- Mean, median, mode, range, and standard deviation conceptually
- The effect of adding or removing a value on centre and spread
- Scatter plots, lines of best fit, and interpreting slope in context
- Interpreting exponential and linear models from data
- Probability from tables and from described situations
- Sample statistics, margin of error, and valid inference
- Evaluating whether a conclusion is supported by a described study

LEARNING OBJECTIVES

- 1 Solve multi-step ratio, rate, and percentage problems accurately and quickly.
- 2 Read any tested display type and answer a contextual question about it.
- 3 Reason about the effect of a data change on centre and spread without computing.
- 4 Judge whether a stated conclusion is supported by the described study design.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers this domain at 90% accuracy within the time budget.
- Interprets displays correctly in 9 of 10 trials.
- Evaluates inference validity correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Percent change problems are computed from the wrong base.	Identify the original quantity explicitly before computing. Successive changes must be applied multiplicatively.
Standard deviation is computed when only comparison was needed.	The SAT asks about relative spread, not calculated values. Reason from the data's clustering.
A conclusion is accepted because the statistics are correct.	Random selection permits generalization; random assignment permits causal claims. The design determines what may be concluded.

TEACHING NOTE

This domain is heavily contextual and reading-dependent. Students whose errors here are comprehension errors rather than mathematical ones should be given reading-focused practice, and the distinction should be made explicitly in the error log.

UNIT 5 • 6-7 LESSONS • THE SMALLEST TESTED DOMAIN

Geometry and Trigonometry

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TOPICS COVERED

- Area and perimeter of plane figures
- Volume and surface area of solids
- Angle relationships: parallel lines, triangles, and polygons
- Triangle similarity and congruence
- The Pythagorean theorem and special right triangles
- Right triangle trigonometry: sine, cosine, tangent
- Complementary angle relationships in trigonometry
- Circles: circumference, area, arcs, sectors, and central and inscribed angles
- The equation of a circle and completing the square
- Radian measure
- Coordinate geometry: distance, midpoint, slope, and lines
- Using the provided reference sheet efficiently

LEARNING OBJECTIVES

- 1 Solve geometry items using the reference sheet efficiently and without memorizing what is provided.
- 2 Apply right triangle trigonometry and special triangles to solve for unknowns.
- 3 Work with circle equations, including converting from general form.
- 4 Recognize when a coordinate approach is faster than a synthetic one.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers geometry and trigonometry items at 88% accuracy within the time budget.
- Uses the reference sheet efficiently rather than reconstructing formulas.

- Selects between coordinate and synthetic approaches appropriately in 8 of 10 items.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Formulas provided on the reference sheet are memorized while those not provided are neglected.	Know exactly what is provided. Trigonometric ratios and the special triangle relationships are not, and must be secure.
Diagrams are assumed to be drawn to scale.	Unless stated, they may not be. Reason from the given information, not from appearance.
Circle equation questions are attempted without completing the square.	Practice the conversion until it is quick; these items are predictable and therefore very learnable.

TEACHING NOTE

This is the smallest domain by weight, which makes it a poor use of time for a student weak in algebra — but it is also the most predictable, which makes it excellent value for a student already strong elsewhere. Sequence accordingly.

UNIT 6 • 6-7 LESSONS PLUS SCHEDULED FULL-LENGTH TESTS

Timed Practice, Adaptive Strategy, and Review

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TOPICS COVERED

- Full-length official practice tests under authentic timed conditions
- Module-level pacing: the per-question time budget and checkpoints
- First module strategy given the adaptive consequence
- Triage: recognizing within fifteen seconds whether to attempt, flag, or skip
- Using the review screen and flagging effectively
- Guessing strategy: no penalty means no blanks
- Item-level error analysis and error log maintenance
- Targeted remediation assigned from the error log
- Managing test anxiety and recovering from a difficult early question
- Test-day logistics: device, Bluebook check-in, permitted materials
- Score interpretation and deciding whether to retest

LEARNING OBJECTIVES

- 1 Complete a full module within the time limit while maintaining accuracy.
- 2 Triage a question within fifteen seconds and act on the decision.
- 3 Maintain an error log and derive the next week's practice from it.
- 4 Execute a stable, rehearsed test-day routine.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Completes modules within time with no unanswered questions in 3 consecutive practice tests.
- Reduces the dominant error category by half from the diagnostic baseline.
- Demonstrates a stable or rising score across the final three practice tests.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Questions are left blank.	There is no penalty for a wrong answer. Every question must be answered, even if by elimination and guess in the final seconds.
Excessive time is spent on a single hard question.	All questions are worth the same. Flag and move; return with the review screen if time permits.
Practice tests are taken without item-level analysis.	The test measures; the analysis teaches. An untaken analysis makes the practice test nearly worthless.

TEACHING NOTE

Practice tests should be spaced roughly every two to three weeks, never weekly. Improvement comes from the remediation between tests, and taking them too frequently leaves no time for the work that actually raises the score.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Diagnostic Practice Test	Before instruction begins	A full official practice test under authentic conditions; produces the item-level baseline that sequences the whole course.
Unit Check	End of each content unit · 25 min	Timed domain-specific sets in SAT format and pacing.
Error Log Review	Every lesson	Ongoing classification of errors into content, strategic, timing, and careless categories; the practice plan is derived from it.
Full Practice Test	Every 2–3 weeks	Official practice tests under authentic timed and digital conditions, followed by item-level analysis.
Final Readiness Assessment	Two weeks before the test date	A full test with a projected score and a final targeted revision plan.

ASSESSMENT	WHEN	WHAT IT MEASURES
Examination Format	Digital SAT · Math section	70 minutes total: two modules of 22 questions, 35 minutes each. Section-adaptive — performance on Module 1 determines the difficulty of Module 2, and routing to the harder module is necessary for the highest scores. Approximately 75% multiple-choice and 25% student-produced response. A calculator is permitted throughout, including the built-in Desmos graphing calculator. A reference sheet of formulas is provided. Questions may be flagged and revisited within a module, but a module cannot be revisited once submitted.

Վարժարանն reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60–75 min	12–16 weeks including practice tests — the most common track.
Intensive	2 lessons/week	8 weeks; suitable when the test date is close and the baseline is already reasonable.
Extended Mastery	1 lesson/week over two testing cycles	24–30 weeks, combining SAT preparation with genuine content remediation for students with real gaps.
Summer Program	3 lessons/week, 6 weeks	Full course before an autumn test date.
Retest Track	1 lesson/week, 6 weeks	Targeted work derived entirely from the previous official score report.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMWORK AND INDEPENDENT PRACTICE

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Forty-five to sixty minutes per lesson: one timed SAT-format set matching the current domain, plus remediation problems assigned from the error log. Every timed set is self-analyzed before the next lesson, with each missed item classified by error type and the correct method written out — not merely the correct answer. Students who complete the analysis consistently improve substantially more than students who complete twice as many problems without it, and this is stated to them explicitly.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Established a diagnostic baseline with item-level error classification.
- ◆ Uses Desmos strategically for graphing, solving, and verification.
- ◆ Enters student-produced responses without formatting losses.
- ◆ Solves all tested algebra content accurately within the time budget.
- ◆ Handles quadratics, exponentials, polynomials, radicals, and rational expressions.
- ◆ Recognizes and exploits equivalent expression structures.
- ◆ Solves ratio, rate, percentage, and unit conversion problems quickly.
- ◆ Interprets all tested data displays and evaluates inference validity.
- ◆ Applies geometry and right triangle trigonometry using the reference sheet efficiently.
- ◆ Completes modules within time with no blanks.
- ◆ Reduced the dominant error category by half from baseline.
- ◆ Achieved the target score on a full official practice test.
- ◆ Completed SAT Mathematics preparation — test ready.

MATERIALS AND RESOURCES

Նյութեր և ռեսուրսներ

- Bluebook application installed on the device the student will use on test day.
- Official College Board practice tests — the primary practice material, used in preference to third-party approximations.
- Desmos graphing calculator practice environment.
- Վարժարանի SAT error log template — the central document of the course.
- Վարժարանի SAT problem bank — approximately 1,800 items tagged by domain, difficulty, and the fastest solution method.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.